

Off Supabase, onto Firebase

The whole backend layer of the app was swapped in one pass — auth, data model, security rules, discovery, and a one-time data migration script. Media storage is the one piece deliberately left on a placeholder.

31 FILES CHANGED	10 NEW FILES	+11,423 INSERTIONS	−3,138 DELETIONS	0 TSC ERRORS	0 SUPABASE REFS LEFT
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THE SWAP

What replaced what

Every Supabase-shaped piece now has a Firebase counterpart. The dependency list shrank: `@supabase/supabase-js`, `@supabase/ssr`, `@supabase/server`, `bcryptjs` and `react-native-url-polyfill` all came out; `firebase ^12.18.0` went in, plus `firebase-admin`, `firebase-tools` and `dotenv` on the dev side for the migration tooling.

CONCERN	WAS	NOW
Sessions	Supabase Auth + bcryptjs	Firebase Auth, AsyncStorage persistence
Database	Postgres tables	Firestore collections
Access control	Row-level security	firestore.rules / storage.rules
Discovery pool	<code>discover_profiles</code> view	<code>profiles/{uid}</code> collection
Client bootstrap	<code>src/services/supabase.ts</code> (deleted)	<code>src/services/firebase.ts</code>
Media	Storage buckets	passthrough placeholder — see below

NEW FILES

The Firebase layer

firebase.ts SHIPPED Config read from <code>EXPO_PUBLIC_FIREBASE_*</code> with a fail-fast check that names the missing keys. Guards for Fast Refresh re-runs. On native it uses <code>initializeAuth</code> with AsyncStorage persistence so a reload no longer signs the user out; on web it falls back to <code>getAuth</code> . Firestore starts with auto-detect long-polling, which React Native's networking stack needs.	firestorePaths.ts SHIPPED One module that owns the entire document layout — every collection and doc reference in the app comes from here, so the schema is readable in a single place and a path can't drift between services.
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discoveryService.ts SHIPPED Reads real members out of <code>profiles</code> , gender-filtered at the query level, and maps them into the existing <code>DiscoverProfile</code> / <code>RishtaListingProfile</code> shapes the screens already expect.	DiscoveryContext.tsx SHIPPED Loads the member pool once per sign-in and hands it to the screens. Falls back to the demo decks when the collection is empty or the fetch fails, and guarantees every card has at least one photo so nothing renders blank.
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DATA MODEL

Public card, private everything else

The split is the point: **one public document per member** that any signed-in user may read, and a private tree underneath that only its owner can touch. That's what removes the need for the old `discover_profiles` view, which only existed to work around row-level security.

PATH	HOLDS	WHO CAN READ
<code>profiles/{uid}</code>	Public member card	Any signed-in member
<code>users/{uid}/private/profile</code>	Email, wali contact	Owner only
<code>users/{uid}/private/verification</code>	CNIC number, media paths	Owner only
<code>users/{uid}/private/notificationPrefs</code>	Notification settings	Owner only
<code>users/{uid}/private/privacyPrefs</code>	Privacy settings	Owner only
<code>users/{uid}/private/dailyLikes</code>	Daily like counter	Owner only
<code>users/{uid}/matches/{id}</code>	Matches	Owner only
<code>users/{uid}/messages/{id}</code>	Chat messages	Owner only
<code>users/{uid}/blocked/{id}</code>	Block list	Owner only
<code>users/{uid}/favorites/{id}</code>	Favourites	Owner only
<code>users/{uid}/viewHistory/{id}</code>	Profile views	Owner only
<code>users/{uid}/notifications/{id}</code>	Notifications	Owner only

Both rule files are written and committed to the repo: `firestore.rules` denies anything not explicitly matched, and `storage.rules` separates `public/{uid}` (readable by members, writable by owner) from `verification/{uid}` (owner-only both ways), with a 25 MB cap and an image/audio/video content-type check.

REWRITES

Services moved over

Every service kept its exported surface, so no screen had to change shape to follow it. The line counts are net churn against the Supabase versions.

MODULE	WHAT IT NOW DOES	Δ LINES
<code>authService.ts</code>	signup, login, logout, <code>getCurrentUser</code> , <code>updateUser</code> , <code>deleteAccount</code> , <code>emailExists</code> — on Firebase Auth, with Firebase error codes mapped to readable messages. Account deletion clears all seven subcollections before removing the user.	858
<code>matchesService.ts</code>	Matches, chat messages and the block list against Firestore.	291

MODULE	WHAT IT NOW DOES	Δ LINES
mediaUpload.ts	Reduced to a documented placeholder (see below).	145
notificationsService.ts	Notification feed and prefs.	94
viewHistoryService.ts	Who-viewed-me history.	72
favoritesService.ts	Favourites.	47
privacyService.ts	Privacy preferences.	36
likeLimitService.ts	Daily like limit.	24
AuthContext.tsx	Session now driven by Firebase's auth state listener.	62
MatchesContext.tsx	Wired to the rewritten matches service.	85

Four screens — Home, Explore, Explore Plus and Profile Detail — stopped importing the mock arrays and now read from `useDiscovery()`. Login lost its email-confirmation branch, which was a Supabase-specific state that doesn't exist here.

TOOLING

The one-time migration script

`scripts/migrate-supabase-to-firebase.mjs` (18 KB, run via `npm run migrate:firebase`) moves existing production data across in one pass: auth users, profiles with their photos and verification records, matches, messages, blocked lists, favourites, view history, notifications, notification prefs, privacy prefs and daily-like counters — copying both storage buckets as it goes.

It takes `--dry-run` and `--default-password=`, and finishes with a tally of what was written, skipped and errored. Credentials come from env vars, and `service-account*.json`, `.firebase/` and `firebase-debug.log` were added to `.gitignore` so a key can't be committed by accident.

COPY

“Dating” became “Friend”

Across all three locale files — English, Roman Urdu and Urdu — the dating-mode wording was reframed as friendship: “*Friends and rishta, in one app*”, *Friend Mode*, and the rishta-transition explainers rewritten to match. Three new strings cover account deletion failures, including the case where Firebase requires a fresh login before it will delete — that one now tells the user to log out and back in, and states plainly that nothing has been deleted yet.

KNOWN GAP

Media storage is not wired

`mediaUpload.ts` PLACEHOLDER

Firestore Cloud Storage requires the paid Blaze plan, and no alternative has been picked yet. Rather than block auth, profiles, discovery and chat, every upload function returns the local device URI unchanged, and that URI is what gets written to Firestore.

- **Photos, intros and chat media render only on the device that picked them.** Another device, or a reinstall, sees a blank card.
- **Removing a photo deletes nothing** — the delete path is a no-op.
- **CNIC and selfie images never leave the device.**

The containment is deliberate: everything that touches media goes through this one module, so swapping in a real backend later is a single-file change — no edits to `authService`, `matchesService`, `discoveryService` or any screen. The Storage rules are already written and ready for the day it's switched on.

ALSO OPEN

Where things stand

- **Everything is uncommitted** on `dev-faizan` — the working tree holds the full migration.
- **The migration script hasn't been run against real data**, not even in dry-run mode.
- **Rules aren't deployed yet** — both files exist locally but no `firebase deploy` has happened.
- **`firestore.indexes.json` is empty.** Composite indexes get added when a query starts needing one.
- **Verified:** `npx tsc --noEmit` passes clean, and no reference to Supabase remains anywhere under `src/`.

SEQUENCE MATTERS HERE

Next

01	Pick a media backend Blaze-plan Storage or a free alternative — it's the only thing blocking real photos, and it decides whether the migration script's file-copy step has a destination.
02	Deploy the rules Firestore and Storage rules need to be live before any real data lands, not after.
03	Dry-run the migration, then run it <code>--dry-run</code> first, read the skipped/errors tally, then go live.
04	Commit The branch is carrying a full backend swap in the working tree.